

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Supplementary Winter Examination – 2024

Course: B. Pharmacy

Subject Name: Medicinal Chemistry-I

Max Marks: 75

Date: 21/12/2024

Semester: IV

Subject code: BP402T

Duration: 03 Hr

Instructions:

1. All questions are compulsory
2. Draw diagrams wherever necessary
3. Figures to right indicates full marks

Q. No. 1 Objective Type Questions (All questions are compulsory) (10X2) = 20

- i) Enlist the Factors affecting drug metabolism.
- ii) Why barbituric acid is inactive and barbiturates are active as sedative and hypnotic?
- iii) Draw any two structures of Solanaceous alkaloids.
- iv) Define the partition coefficient and give its importance in relation to biological activity.
- v) Write sulphonation as phase II metabolism reaction.
- vi) Write a note on Cholinesterase reactivator.
- vii) Give the synthesis of Ketamine hydrochloride.
- viii) Why phenytoin does not give sedation action?
- ix) Draw the structure of narcotic antagonist with their MOA.
- x) Define and give importance of Bioisosterism.

Q. No. 2 Long Answer Questions (Answer 2 out of 3) (10X2) = 20

- i) Define metabolism, give its significance and explain oxidation as Phase I metabolism reaction.
- ii) Define and classify parasympathomimetic agents and explain in detail SAR and MOA Acetylcholine.
- iii) Define and classify sedatives and hypnotics and explain in detail SAR and MOA of Barbiturates.

Q. No. 3 Short Answer Questions (Answer 7 out of 9) (5X7) = 35

- i) Write a note on Ionization and Solubility.
- ii) Enlist Phase II metabolism reaction and explain glucuronidation reaction.
- iii) Define and classify sympathomimetic agents.
- iv) Explain in detail SAR and MOA of β -blocker.
- v) Define and classify Anticonvulsants.
- vi) Explain in detail SAR of Phenothiazines derivative.
- vii) Define and classify General anesthetics.
- viii) Define and classify Anti-inflammatory agents.
- ix) Explain SAR of Morphine analogues.

END OF THE PAPER